MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCECT203 Wireless Technology (Minor for Others)

Duration: 2 hours **Maximum Marks: 35**

Practical Examination - Model Question Paper

Instructions:

- Demonstrate the given task.
- A viva based on the practical modules.
- A lab report covering the practical modules.

Evaluation Criteria:

- Viva : 10 marks
- Lab record: 10 marks
- Demonstration: 15 marks

Tasks: *(Choose any one from the list)*

1. Implement a simple IR LED transmitter and receiver circuit. Observe and record the output waveform.
2. Simulate a basic LC resonant circuit using LTspice (or equivalent). Identify the resonant frequency from the output.
3. Simulate an AM modulator circuit. Apply a message signal and carrier signal. Observe and explain the modulated waveform.
4. Implement a simple diode-based envelope detector. Apply an AM signal as input. Observe the recovered signal at the output.
5. Design and simulate a basic RC low-pass or band-pass filter. Plot the frequency response. Identify the cut-off frequency.
6. Use RTL-SDR to observe nearby RF signals. Identify at least one signal band. Record frequency and signal strength.
7. Control an LED using Wi-Fi (ESP32/NodeMCU). Demonstrate ON/OFF operation through a simple interface. Explain the working principle.
8. Establish wireless communication between two nodes using NRF24L01. Transmit a simple data value. Verify successful reception.
9. Set up RF transmitter and receiver modules (FS1000A & XY-MK). Transmit a digital signal. Observe received output.
10. Interface HC-05 Bluetooth module with Arduino/ESP32. Send data from a mobile device to the microcontroller. Display or act on received data.
11. Interface a temperature / humidity / potentiometer sensor. Transmit sensor data wirelessly. Display the received data.
12. Interface PN532 NFC module with a microcontroller. Read an NFC card/tag. Display the card ID.
13. Demonstrate the basic block diagram of a cellular IoT system using A7672S. Explain the role of LTE in IoT communication.
14. Transmit a wireless signal using any available module (IR / RF / Wi-Fi). Measure the signal response at different distances. Comment on the effect of distance on signal strength.
15. Simulate a simple RLC circuit in LTspice. Perform frequency sweep analysis. Identify passband and stopband regions.
16. Transmit sensor or control data wirelessly using ESP32 / Arduino. Display the received data on Serial Monitor. Explain the data flow from transmitter to receiver.